

Explanation of the Mode II to Akantu Parameter Conversion Function

Gauss T. Chang[†]

[†]*Department of Civil Engineering, R14521220@ntu.edu.tw*

November 19, 2025

I Introduction

In Akantu, cohesive elements are governed by a mixed-mode failure criterion. However, in many experimental or theoretical studies, material properties are obtained in pure Mode II, such as the critical shear strength τ_c and the Mode II fracture energy G_{IIc} . Akantu, in contrast, requires the following material parameters for its cohesive zone model:

- σ_c : the critical cohesive strength associated with the mixed-mode effective stress criterion,
- G_c : the fracture energy,
- β : the mode-mixity parameter controlling the contribution of shear in the failure criterion.

The Python function provided performs the conversion from the physically known Mode II quantities (τ_c, G_{IIc}) into the parameters required by Akantu (σ_c, G_c, β) .

II Akantu Effective Stress Criterion

Akantú defines an effective cohesive stress as

$$\sigma_{\text{eff}} = \sqrt{\sigma_n^2 + \frac{\tau^2}{\beta^2}}, \quad (1)$$

where σ_n is the normal traction and τ is the shear traction on the cohesive interface. Cohesive failure initiates when

$$\sigma_{\text{eff}} > \sigma_c. \quad (2)$$

III Specialization to Pure Mode II Loading

For a pure Mode II simulation, the opening stress is negligible, $\sigma_n \approx 0$, so that the effective stress reduces to

$$\sigma_{\text{eff}} = \frac{\tau}{\beta}. \quad (3)$$

At the onset of failure,

$$\sigma_{\text{eff}} = \sigma_c, \quad (4)$$

which implies:

$$\frac{\tau_c}{\beta} = \sigma_c \quad \Longleftrightarrow \quad \sigma_c = \frac{\tau_c}{\beta}. \quad (5)$$

Thus, the Akantu parameter σ_c is directly computed from the desired Mode II shear strength τ_c and the chosen mode-mixity parameter β .

IV Fracture Energy and Critical Shear Displacement

In many cohesive zone formulations, including the bilinear model, the Mode II fracture energy is related to the critical shear displacement jump γ_c through

$$G_{IIc} = \frac{1}{2} \tau_c \gamma_c. \quad (6)$$

Solving for γ_c yields

$$\gamma_c = \frac{2G_{IIc}}{\tau_c}. \quad (7)$$

While Akantu does not require γ_c as an input, it is computed in the Python function for consistency checking and for interpreting the physical scale of the cohesive separation.

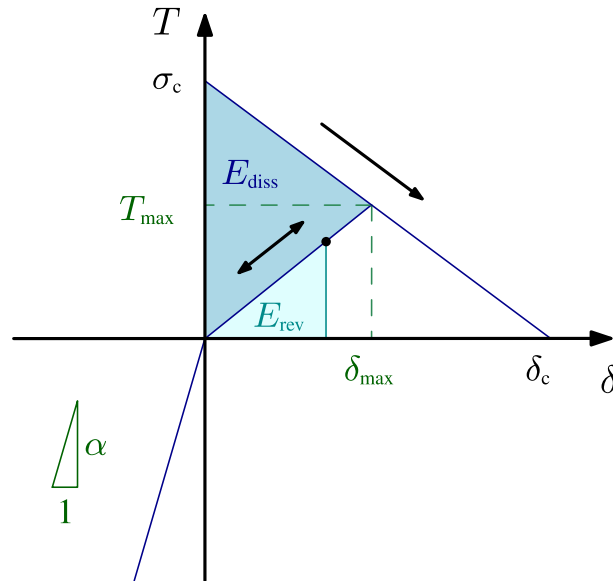


Figure 1: linear cohesive law.

V Function Summary and Interpretation

The Python function converts Mode II parameters to Akantu input parameters:

- It computes $\sigma_c = \tau_c/\beta$, which is required by Akantu’s effective stress criterion.
- It sets $G_c = G_{IIc}$ for a pure Mode II analysis.
- It computes $\gamma_c = 2G_{IIc}/\tau_c$ for validation purposes.
- It returns $(\sigma_c, G_c, \beta, \gamma_c)$ to be used in the material definition.

The function therefore allows a user to select a small value of β to suppress Mode I failure and promote Mode II behavior, while preserving the physically correct Mode II fracture properties.

VI Conclusion

The conversion procedure ensures that a user who possesses Mode II cohesive properties can correctly construct an Akantu cohesive material model. The relationship $\sigma_c = \tau_c/\beta$ is essential for obtaining the desired shear-dominated failure behavior, especially when using a very small value of β to suppress Mode I failure.